Capstone Project Proposal

University Student Competition System

Platform: Web & Mobile

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1 Introduction

The University Student Competition System (USCS) is a comprehensive web and mobile platform designed to streamline the organization, participation, and evaluation of competitions within a university. The system serves four key user roles Students, Organizers, Judges, and Administrators each with dedicated tools and permissions.

The platform consolidates all university competitions into one unified digital system, simplifying registration, judging, and result management. It also integrates AI-powered recommendations to suggest suitable competitions to students based on their interests and participation history, ensuring accessibility, transparency, and engagement.

2 Problem Statement

Students often miss competition opportunities due to fragmented announcements and communication.

Organizers face challenges managing registration, scheduling, and communication manually.

Judges struggle with time-consuming and inconsistent scoring methods.

Administrators lack centralized oversight of competition data and participation trends.

The absence of a unified platform limits visibility, efficiency, and fairness in competition management.

3 Project Objectives

- Develop a centralized platform to manage all competitions within the university efficiently.
- Provide role-based dashboards for students, organizers, judges, and administrators.
- Automate competition registration, evaluation, and result publication.
- Integrate AI-driven recommendations to match students with competitions that suit their skills and interests.
- Ensure transparent scoring and ranking through digital evaluation and result tracking.
- Enable real-time communication between participants, judges, and organizers.
- Ensure security, scalability, and user-friendly access across web and mobile platforms.

4 Scope of the Project

- Multi-role access and management: Dedicated dashboards for students, organizers, judges, and administrators, each with defined roles and permissions.
- Competition lifecycle management: End-to-end digital handling of event creation, registration, scheduling, evaluation, and result publishing for all competitions within the university.
- AI-based recommendation engine: Personalized suggestions for students based on interests, academic background, and previous participation.
- Transparent judging and scoring: Digital evaluation forms, automated score calculation, and real-time leaderboard updates.
- Real-time communication and notifications: In-app chat, alerts, and updates for announcements, schedule changes, and results.
- Secure authentication and data privacy: Role-based access control, encrypted data storage, and compliance with institutional security standards.
- Scalable backend architecture: Built using modern frameworks (e.g., Node.js/Express) and databases (e.g., MySQL/PostgreSQL) to handle large user traffic and real-time operations effectively.
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5 Methodology

- Requirement Analysis: Conduct surveys and interviews with students, organizers, and faculty to gather detailed requirements and understand user expectations.
- System Design: Create architecture for the frontend, backend, and AI modules. Design data flow diagrams, ER diagrams, and user interface mockups for all user roles.
- Development:
  - Frontend: React.js for web and Flutter for mobile.
  - Backend: Node.js/Express with a MySQL or PostgreSQL database.
  - AI Module: Machine learning-based recommendation model for suggesting competitions.
  - Authentication: JWT-based secure login and role-based authorization.
- Testing: Perform unit testing, integration testing, and user acceptance testing to ensure system reliability and ease of use.
- Deployment: Deploy the system to a secure cloud platform, ensuring scalability, uptime, and data protection.

6 Expected Outcomes

- A unified digital platform that manages all competitions within the university efficiently, reducing manual workload and communication gaps.
- Simplified workflows for students, organizers, judges, and administrators, ensuring clarity and collaboration.
- Automated registration, evaluation, and scoring systems that improve transparency and fairness.
- Increased student engagement through AI-powered personalized competition recommendations.
- Reliable analytics and reporting tools for administrators to track participation, performance, and trends.
- A scalable and secure digital foundation designed for future integration with other university systems such as student portals or academic management tools